

Day-7

DATE:13-08-25

NAME: NANCY M

Program.cs:

```
namespace Day8task
{
    class Program
    {
        public static void Main(string[] args)
        {

            PaintBrush brush = new PaintBrush();
            SprayGun sprayGun = new SprayGun();

            Painter john = new Painter("John", brush);
            Painter alice = new Painter("Alice", sprayGun);

            john.StartPainting();
            alice.StartPainting();
        }
    }
}
```

ITool.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Day8task
{
    public interface ITool
```

```
{  
    string Use();  
}  
}
```

Painter.cs:

```
using System;  
using System.Collections.Generic;  
using System.Linq;  
using System.Text;  
using System.Threading.Tasks;  
  
namespace Day8task  
{  
    public class Painter  
    {  
  
        private readonly string _name;  
        private readonly ITool _tool;  
  
        public Painter(string name, ITool tool)  
        {  
            _name = name;  
            _tool = tool;  
        }  
  
        public void StartPainting()  
        {  
            Console.WriteLine($"{_name} is {_tool.Use()}");  
        }  
    }  
}
```

PaintBrush.cs:

```
using System;  
using System.Collections.Generic;  
using System.Linq;
```

```
using System.Text;
using System.Threading.Tasks;

namespace Day8task
{
    public class PaintBrush : ITool
    {
        public string Use() => "Painting with PaintBrush";
    }
}
```

SprayGun.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Day8task
{
    public class SprayGun : ITool
    {
        public string Use() => "Painting with SprayGun";
    }
}
```

UnitTest1.cs:

```
using Day8task;

namespace PainterTest
{
    public class Tests
    {
        [Test]
        public void Painter_WithPaintBrush_ShouldReturnCorrectMessage()
        {
            var painter = new Painter("John", new PaintBrush());
            using var sw = new StringWriter();
        }
    }
}
```

```
    Console.SetOut(sw);

    painter.StartPainting();

    Assert.That(sw.ToString().Trim(), Is.EqualTo("John is Painting with PaintBrush"));
}
```

```
[Test]
public void Painter_WithSprayGun_ShouldReturnCorrectMessage()
{
    var painter = new Painter("Alice", new SprayGun());
    using var sw = new StringWriter();
    Console.SetOut(sw);

    painter.StartPainting();

    Assert.That(sw.ToString().Trim(), Is.EqualTo("Alice is Painting with SprayGun"));
}
```

```
[Test]
public void PaintBrush_Use_ShouldReturnCorrectString()
{
    var brush = new PaintBrush();
    Assert.That(brush.Use(), Is.EqualTo("Painting with PaintBrush"));
}
```

```
[Test]
public void SprayGun_Use_ShouldReturnCorrectString()
{
    var gun = new SprayGun();
    Assert.That(gun.Use(), Is.EqualTo("Painting with SprayGun"));
}
```

```
[Test]
public void Painter_CanSwitchToDifferentTool()
{
    var painter = new Painter("Mike", new PaintBrush());
    using var sw1 = new StringWriter();
    Console.SetOut(sw1);
    painter.StartPainting();
}
```

```

    Assert.That(sw1.ToString().Trim(), Does.Contain("PaintBrush"));

    painter = new Painter("Mike", new SprayGun());
    using var sw2 = new StringWriter();
    Console.SetOut(sw2);
    painter.StartPainting();
    Assert.That(sw2.ToString().Trim(), Does.Contain("SprayGun"));
}

```

```

[Test]
public void Painter_Output_ShouldContainPainterName()
{
    var painter = new Painter("Sam", new PaintBrush());
    using var sw = new StringWriter();
    Console.SetOut(sw);

    painter.StartPainting();

    Assert.That(sw.ToString(), Does.Contain("Sam"));
}

```

```

[Test]
public void Tools_ShouldImplementIToolInterface()
{
    Assert.That(typeof(ITool).IsAssignableFrom(typeof(PaintBrush)), Is.True);
    Assert.That(typeof(ITool).IsAssignableFrom(typeof(SprayGun)), Is.True);
}

```

```

[Test]
public void Painter_Output_ShouldNotBeEmpty()
{
    var painter = new Painter("John", new PaintBrush());
    using var sw = new StringWriter();
    Console.SetOut(sw);

    painter.StartPainting();

    Assert.That(sw.ToString().Trim(), Is.Not.Empty);
}

```

```

[Test]
public void MultiplePainters_ShouldWorkIndependently()
{
    var painter1 = new Painter("A", new PaintBrush());
    var painter2 = new Painter("B", new SprayGun());

    using var sw1 = new StringWriter();
    Console.SetOut(sw1);
    painter1.StartPainting();
    var output1 = sw1.ToString();

    using var sw2 = new StringWriter();
    Console.SetOut(sw2);
    painter2.StartPainting();
    var output2 = sw2.ToString();

    Assert.That(output1, Does.Contain("PaintBrush"));
    Assert.That(output2, Does.Contain("SprayGun"));
}

```

```

[Test]
public void Painter_Output_ShouldBeInCorrectFormat()
{
    var painter = new Painter("Tom", new PaintBrush());
    using var sw = new StringWriter();
    Console.SetOut(sw);

    painter.StartPainting();

    var output = sw.ToString().Trim();
    StringAssert.IsMatch(@"^[A-Za-z]+ is Painting with [A-Za-z]+$", output);
}
}
}

```

```
Microsoft Visual Studio Debug Console
John is Painting with PaintBrush
Alice is Painting with SprayGun
C:\Users\nancy.m\source\repos\Day8task\Day8task\bin\Debug\net8.0\Day8task.exe (process 20080) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

Test Explorer

Build succeeded 0 Warnings 0 Errors

Test	Duration	Traits	Error Message
PainterTest (10)	35 ms		
PainterTest (10)	35 ms		
Tests (10)	35 ms		
MultiplePainters_ShouldWorkIn...	8 ms		
PaintBrush_Use_ShouldReturnC...	4 ms		
Painter_CanSwitchToDifferentTool	< 1 ms		
Painter_Output_ShouldBeInCorr...	9 ms		
Painter_Output_ShouldContain...	< 1 ms		
Painter_Output_ShouldNotBeE...	6 ms		
Painter_WithPaintBrush_Should...	7 ms		
Painter_WithSprayGun_ShouldR...	< 1 ms		
SprayGun_Use_ShouldReturnCo...	< 1 ms		
Tools_ShouldImplementIToolInt...	1 ms		

Group Summary

PainterTest

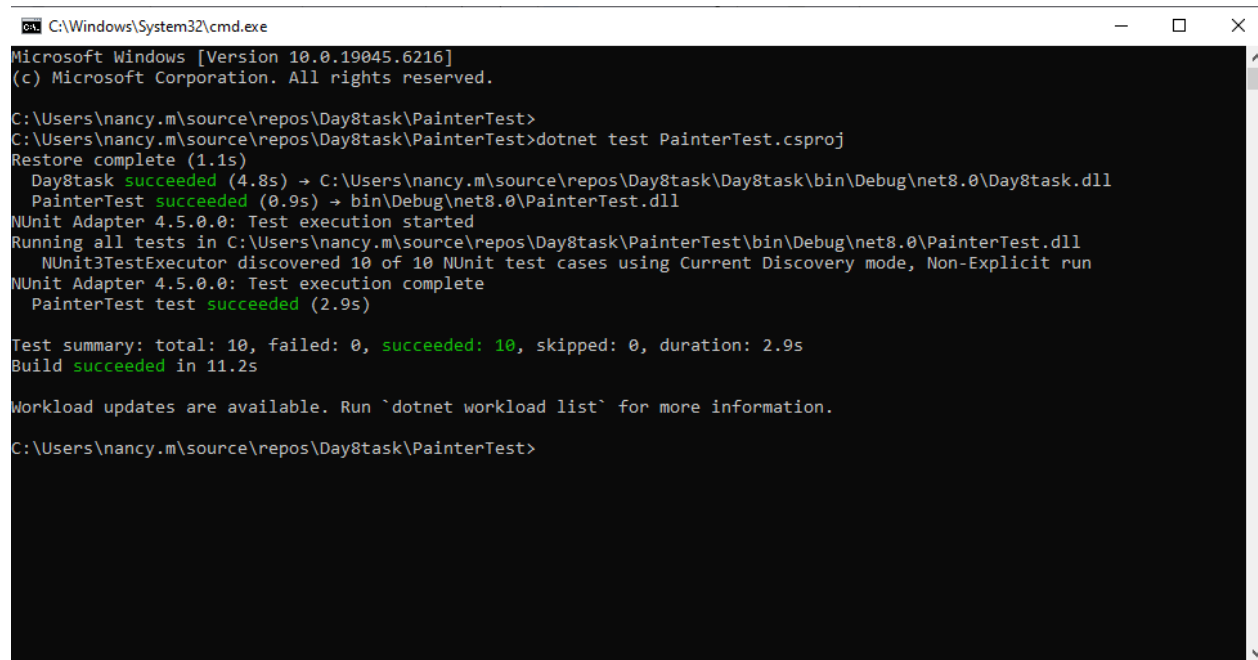
Tests in group: 10

Total Duration: 35 ms

Outcomes

10 Passed

2.1 run the tests using both visual studio and command line



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6216]
(c) Microsoft Corporation. All rights reserved.

C:\Users\nancy.m\source\repos\Day8task\PainterTest>
C:\Users\nancy.m\source\repos\Day8task\PainterTest>dotnet test PainterTest.csproj
Restore complete (1.1s)
  Day8task succeeded (4.8s) -> C:\Users\nancy.m\source\repos\Day8task\Day8task\bin\Debug\net8.0\Day8task.dll
  PainterTest succeeded (0.9s) -> bin\Debug\net8.0\PainterTest.dll
NUnit Adapter 4.5.0.0: Test execution started
Running all tests in C:\Users\nancy.m\source\repos\Day8task\PainterTest\bin\Debug\net8.0\PainterTest.dll
  NUnit3TestExecutor discovered 10 of 10 NUnit test cases using Current Discovery mode, Non-Explicit run
NUnit Adapter 4.5.0.0: Test execution complete
  PainterTest test succeeded (2.9s)

Test summary: total: 10, failed: 0, succeeded: 10, skipped: 0, duration: 2.9s
Build succeeded in 11.2s

Workload updates are available. Run `dotnet workload list` for more information.

C:\Users\nancy.m\source\repos\Day8task\PainterTest>
```